$l_n = 0,66$ $\quad l_c = 0,46$

$m_c = 1,92$ $\quad m_p = 0,720$

$v_{s,0} \approx 7,5$

$$k_t = -\frac{l_2}{v_{s,0} \cdot l_n} = -\frac{l_c \cdot m_c \cdot g - 2 l_n m_p g}{v_{s,0} \cdot l_n}$$

$$k_4 = -4,366$$

$$v_{s,0} = \frac{(2 l_n m_p - l_c m_c) g}{l_n k_t}$$

$$k_t = \frac{2 l_n m_p - l_c m_c}{l_n v_{s,0}} \approx 0,133178$$